Tanachai Vilainate

Research Assistant | Carbon Capture

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ABOUT ME

Environmental Engineering graduate with hands-on experience in data analysis, KPI tracking, and reporting across industrial research projects. Skilled in Excel, SQL, and Power BI, with a track record of turning operational data into actionable insights, with growing expertise in machine learning and AI-assisted analytics

SKILLS

- Excel (Power Query, Power Pivot, Advanced)
- SQL (Joins, Subqueries, CTE, SET Operations)
- Python (Pandas, NumPy, Scikit-learn, Matplotlib)
- Machine Learning (Regression, Classification, Model Evaluation)
- Power BI
- Financial Statement Reading
- LLM-assisted workflows (Claude, ChatGPT)

CERTIFICATES

- Microsoft Office by Mahidol
- Excel Skills for Business Advanced
- ESG 101 Sustainability
- SQL for Data Science
- Soft Skills Training for Employment

LANGUAGES

- Thai (Native) | English (TOEIC 700)

EXPERIENCES

Asian Institute of Technology, Bangchak, and Mitico
September 2025 - May 2026

Department of Environmental Engineering and Management
Carbon Capture

- Acted as a focal point between AIT, Mitico, and Bangchak for technical updates, troubleshooting, and scheduling.
- Oversaw daily operation of the carbon capture unit.
- Tracked 4 KPIs across multi-run experiments; findings informed sorbent selection and reduced operational costs.
- Analyzed the data of CO_2 Captured and Recovered using Excel, significantly detected Gen 2 sorbent inefficiency, determined it as the root cause of operational issues, and drove cost savings by discontinuing its production.
- Prepared and delivered monthly progress reports to 3 stakeholder organizations (AIT, Bangchak, Mitico).

Mass Rapid Transit Authority of Thailand
April 2023 - June 2023

Internship

- Monitoring environmental issues arising from the construction of MRT station and collecting the field data.
- Preparing reports from the collected data and verifying accuracy against the original EIA document.
- Presenting to the supervisor to identify problems and propose solutions.

PROJECTS

Coffee Shop Daily Revenue Prediction

Kaggle datasets

- Performed end-to-end analysis (ETL, dashboard, reporting) on Kaggle coffee shop dataset using Claude Cowork (Excel, and Python).
- Built a multiple linear regression model (Python, pandas, scikit-learn) to predict daily revenue, achieving R^2 of 0.9490 with Random Forest Model.
- Performed feature selection and statistical significance testing to identify 3 key drivers
- Found 3 significant impact factors; each $1 increase in Average Order Value contributes $243 to daily revenue.

EDUCATION

Kasetsart University
Bangkok, Thailand April 2024

- Bachelor of Engineering in Environmental Engineering, 3.30 GPA (2nd class honours)